CAUSAL DISSECTION OF SCIENTIFIC AGENTS: BREAKS FROM EVIDENCE TO ACTION

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Paper under double-blind review

ABSTRACT

Experimental success does not reveal whether a scientific agent learned the right model. We use ChemWorld as a controlled causal probe: physics, interface, resources, and stochastic identity stay fixed while the initial model is opaque, aligned, or misspecified at entity, parametric, or structural loci. Across 135 sessions nested within 45 independent task–world clusters, search patterns differed across arms and prediction error fell in every arm and locus, yet all three selective-correction criteria failed. After matched one-pair partition packets, mean error fell from 0.2255–0.3392 to 0.0060–0.0074 and exact 1.75-law expression was 0/5 in both models, but this one-pair free-text surface admits an exact linear/power alias. On separate matched 30-cell reference-fitter-identifiable surfaces, failure-aware joint recovery was 0/30 for DeepSeek-v4-flash (17 completed, 13 schema failures) and 5/30 for GPT-5.6-sol, with no useful-gain success among 18 scheduled opportunities per model. On fully scheduled 135-cell surfaces, the 129/135 available GPT-5.6-sol laws had lower MAE than the 135/135 DeepSeek laws (0.175 versus 0.237), yet blind gain remained approximately zero in both. In an oracle-free four-condition development successor, the all-scheduled autonomy-minus-no-evidence estimate was -0.091 for DeepSeek-v4-flash and +0.110 for GPT-5.6-sol; both intervals crossed zero. In the DeepSeek longitudinal cohort, re-executing the last-available laws from all 45 cells over the frozen candidate packets yielded 0/45 law-implied versus 11/45 participant Top-1 choices. Finally, a ranking control passed 7/8 fresh units while selecting Top-1 in 1/8, whereas another failed the rank gate but selected Top-1 with zero regret. Numerical revision, structural identification, executable compression, decision transfer, and evaluator validity are distinct transitions rather than one score.

1 INTRODUCTION

Autonomous experimental agents are increasingly judged by whether they discover high-performing molecules or reactions. Yet an endpoint benchmark cannot distinguish whether an agent inherited a fortunate prior, navigated a productive search corridor, adopted a phenomenological interpolation, or genuinely revised its scientific understanding from evidence. We call this ambiguity endpoint underdetermination. The relevant question is not only whether an agent finds a good experiment, but what that experiment changes in its model of the world.

This distinction matters as language-model agents plan syntheses, call chemistry tools, operate instruments, and enter self-driving laboratory workflows (Boiko et al., 2023; Bran et al., 2024; Szymanski et al., 2023; Darvish et al., 2025; Song et al., 2025; Vriza et al., 2026). Interactive environments increasingly test repeated cycles of hypothesis, intervention, and inference (Jansen et al., 2024; Gandhi et al., 2025; Duan et al., 2025; Zheng et al., 2026; Yang et al., 2026; Batzoglou, 2026). Yet pretrained knowledge, prompt-provided information, experiment selection, endpoint optimization, and verbal explanation are usually observed together. Consequently, a high score need not identify which transformation succeeded, and a low score need not identify which transformation failed.

We make those transformations observable by using ChemWorld (Qiu et al., 2026) as a controlled causal probe. The same executable world, interface, resources, and stochastic identity are held fixed while its initial model is made opaque, aligned, or misspecified at a declared entity, parametric, or

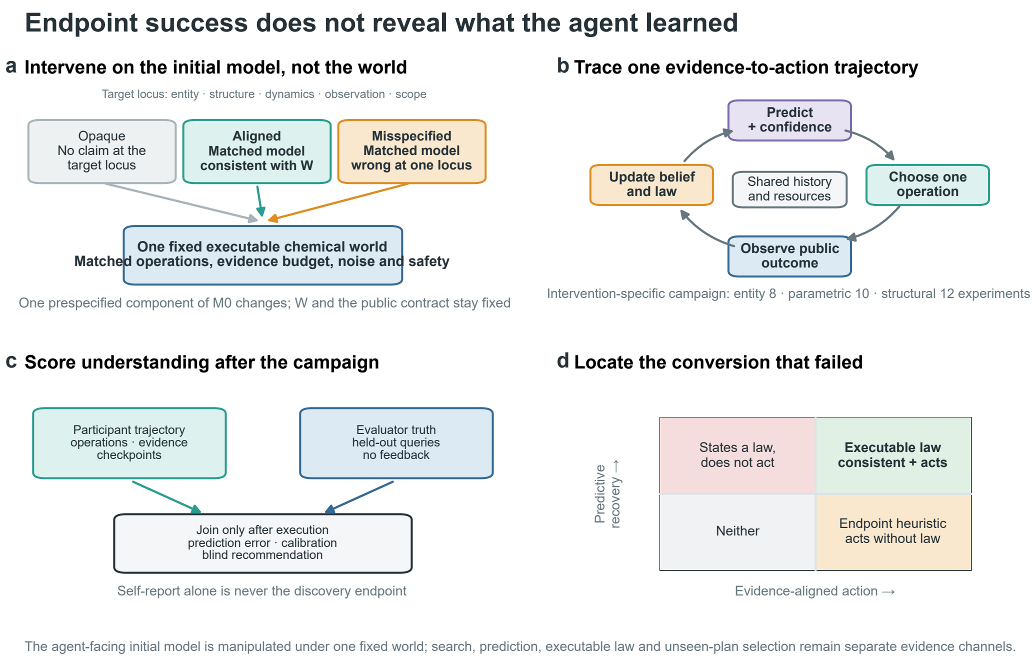

Figure 1: **Endpoint success underdetermines scientific competence.** The same outcome can arise from a correct prior, local search, or evidence-driven model correction. ChemWorld intervenes on the initial model while holding the executable world fixed, then evaluates the transitions from evidence to predictions, executable laws, and unseen actions.

structural locus. Persistent agents then interact through a transactional laboratory interface. Evaluator-owned counterfactual queries, matched contradictory evidence, executable-law assays, and unseen complete ActionPlans separately measure search, prediction, structural identification, compression, and decision transfer.

Our contribution is a transition map rather than a scalar leaderboard. Its causal scope is explicit: initial-model assignment manipulates the participant-facing starting model, while matched counterevidence probes conditional post-packet response. Law–action analyses are descriptive associations, the four-condition successor estimates development-stage information strategies with system failures retained, and the oracle studies diagnose evaluator controls. Across these layers, four breaks emerge: search patterns differ across initial-model arms without establishing selective repair; post-packet numerical convergence does not certify structure, and recovery remains sparse on an identifiable control; executable laws can lose information present in explicit predictions; and neither law error nor rank qualification is a sufficient proxy for action validity.

2 Environment and Evaluation

2.1 ChemWorld as a controlled causal probe

ChemWorld separates three objects. The executable world contains hidden state, transition and measurement laws, resources, and terminal assays. The public task exposes typed operations, instruments, observations, and an initial model. The evaluator owns held-out truth, replay, and scoring. For world w and initial-model arm a, the hidden transition process remains fixed while the supplied model $M_{0,a}$ changes. The aligned model is correct at one declared locus, the misspecified model is explicit but wrong at that locus, and the opaque arm withholds the corresponding structure. The design changes participant-facing information while holding world physics and the available operation and measurement semantics fixed.

Actions are transactions, not free-form prose. The host validates a typed request, checks resources and preconditions, commits the state transition, and returns a public observation or a structured rejection. Failed actions and discards remain in the trajectory. Every submitted action, commit, rollback, measurement, assay, and resource delta is recorded for exact replay. These semantics matter: an action recommendation is scientific evidence only when the evaluator can execute the same complete plan without filling in hidden workflow choices.

2.2 Five-stage capability chain

We distinguish five downstream outcomes. **Evidence acquisition** asks which informative experiments are selected and observed. **Numerical revision** asks whether held-out counterfactual prediction error falls. **Structural identification** asks whether the agent rejects a false form and recovers the registered relation. **Executable compression** asks whether a typed law preserves the information in conditional predictions. **Decision transfer** asks whether the final state selects a good previously unseen plan. A sixth requirement, **evaluator validity**, asks whether the control and metric actually measure the decision estimand. Success at one stage is neither defined as nor assumed to imply success at the next.

At five checkpoints per session, the participant reports initial-model reliability, predictions, uncertainty, evidence references, an executable law, and the next experimental intent. The evaluator executes prespecified counterfactual query sets independently of the participant. The DeepSeek-v4-flash surface scores 675/675 checkpoints from 420/420 truth executions; the GPT-5.6-sol surface scores 669/675 checkpoints from its own 420/420 truth executions. The evaluator later runs each available final typed law on the same coordinates. Paired blind replay evaluates the final recommendation against the observed incumbent, while a separate longitudinal assay reveals eight outcome-hidden complete ActionPlans only after autonomous exploration has ended.

3 Experimental programme

The prospective programme is layer-stratified. Entity interventions cover five task families and five independently selected public worlds per task. Parametric and structural blocks each cover two validated task families and five worlds. Every task–world cluster contains opaque, aligned, and misspecified arms, yielding 135 separate sessions nested within 45 independent task–world clusters. Campaign length is locus-specific: eight, ten, or twelve complete experiments, with five checkpoints in every session. Exploratory, validation, prospective, matched-evidence, and open-action worlds remain separated. We abbreviate the prospective cohort as C2, the matched partition-packet diagnostic as B2, its typed-law/action control as B3, and the entity/parametric/structural loci as A-E/A-P/A-S.

The primary contrast tests selective evidence-driven correction. If $E^{(\ell)}_{a,k}$ is held-out error for arm a, checkpoint k, and locus ℓ, then

$$C_\ell = \left(E^{(\ell)}_{\mathrm{mis},0} - E^{(\ell)}_{\mathrm{mis},K}\right) - \left(E^{(\ell)}_{\mathrm{aligned},0} - E^{(\ell)}_{\mathrm{aligned},K}\right).$$

Success requires greater correction in the misspecified arm, improvement of that arm, and no material deterioration of the aligned arm. Loci are decided separately; unlike intervention semantics are not pooled. Failed cells stay in the scheduled denominator; confirmatory correction gates use adverse bounds, while last-observation and zero-improvement imputations are sensitivities. Only infrastructure failures without a persisted trajectory can resume under a fixed attempt cap.

Matched-evidence sessions use cloned worlds and give every arm the same counterevidence after a pre-response. They reveal conditional post-packet updating but, without a turn-matched no-packet control, do not identify a pure evidence-packet effect. The longitudinal action matrix separately contains three tasks, five worlds, and three arms (45 scheduled cells). After 12 autonomous experiments, each agent ranks eight new plans; regret and Top-1 are primary action readouts, while complete-rank correlation and law adequacy are diagnostics. Table~1 keeps these layers and their claim boundaries explicit.

Provenance is block-specific: C2 and B3 have matched scheduled DeepSeek-v4-flash-high and GPT-5.6-sol-medium surfaces; matched evidence adds complete denominators and a DeepSeek-low

Table 1: **Executed evidence layers and claim boundaries.** A completed work package may contain a retained scientific rejection; unstarted units are not silently removed.

Layer	Units	Execution	Supported role
Prospective C2	270 scheduled	DeepSeek 121 complete; GPT 126 complete	Search, prediction, law, incumbent replay
Matched evidence	75 sessions	DeepSeek high + GPT medium (60); DeepSeek low (15)	Conditional numerical–exact-law-expression dissociation
Identifiable-law B3	60 scheduled	DeepSeek 17+13 failures; GPT 30	Structural recovery and action bridge
Action assays	45 cells + 360 slots	Open action + four conditions/model	Descriptive and failure-aware action transfer
96-query control	15 planned	Provider-free; 8 attempted	Retained pre-participant rejection
320-query control	7 exposed + 15 fresh	Provider-free; 7 pass + 1 fresh	Construction repair versus prospective rejection
Gate alignment	16 unit versions	Provider-free; 0 new execution	Rank validity versus action validity

B2 reasoning-budget ablation; the four-condition successor retains model-specific donor eligibility. Provider-free controls and model configurations are never pooled.

4 Results: from priors to laws

4.1 Priors scaffold search without selective correction

All 135 scheduled sessions produced final records. Participants completed 1,243/1,260 planned experiments; 121 sessions met operational eligibility. The denominator retains 26 discarded lifecycles, 13 resource-ledger rejections, and all right-censored cells. Every session submitted five checkpoints, providing 6,300 counterfactual predictions and 24,300 query–metric values.

Arm assignment was reflected in behavior. The first complete recipe differed between aligned and misspecified cells in 45/45 matched clusters, between opaque and aligned in 45/45, and between opaque and misspecified in 44/45. This is a manipulation check rather than a causal effect estimate because there are no repeated same-arm sessions. Search continued after the first proposal: 91.2% of completed experiments used a unique recipe, 84.4% of session optima appeared after the midpoint, and 32.6% appeared in the last completed experiment.

Correct-prior utility was task dependent. In entity-level partition, the aligned arm showed a +0.200 best-endpoint advantage over the misspecified arm in 5/5 worlds. In structural crystallization, a +0.141 first-experiment head start narrowed to +0.055 as the disadvantaged arm explored. In structural partition, aligned and misspecified descriptions both helped relative to opaque identifiers while differing little from one another. The supplied-model arms therefore occupied different search landscapes, without identifying a stochastic participant effect or imposing one endpoint ordering.

Prediction error nevertheless fell on average in every arm at every locus. Reductions for opaque, aligned, and misspecified cells were 0.111/0.097/0.097 at the entity locus, 0.090/0.033/0.065 at the parametric locus, and 0.219/0.228/0.221 at the structural locus. The stricter selective-correction contrasts were -0.214 for entity ($p = 0.990$), +0.033 for parametric ($p = 0.079$), and -0.224 for structural ($p = 1.000$); none passed. General predictive learning did not become preferential correction of a wrong starting model (Fig.~2).

4.2 Matched packets expose numerical–expression dissociation; B3 tests structure

Free discovery cannot distinguish failure to seek diagnostic evidence from failure to use it. After a decisive parametric packet, all five misspecified summaries rejected the supplied high-potential direction and recovered peak-and-collapse. This packet-plus-turn response does not isolate the packet.

After the same B2 phase-process packet, opaque/aligned/misspecified mean errors fell from 0.2255/0.2736/0.3392 to 0.0074/0.0060/0.0071. The update contrast was positive in 3/5 worlds (exact one-sided $p = 0.125$), and retrospective coding found 0/5 misspecified summaries expressing

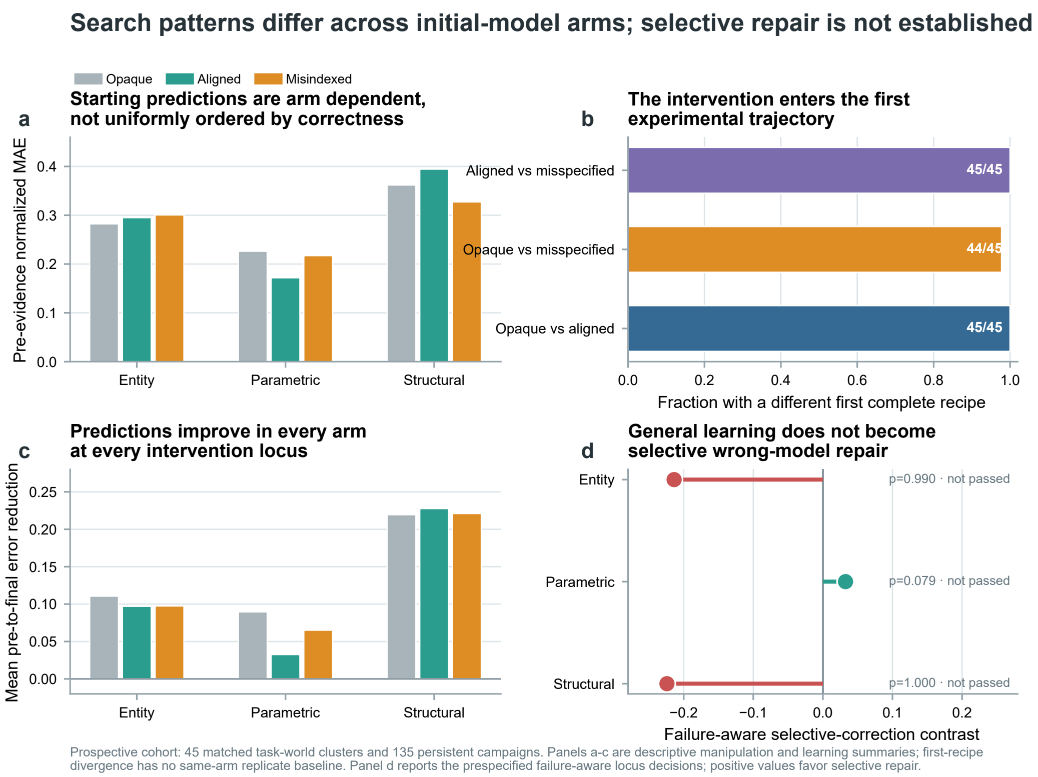

Figure 2: **Search patterns differ across initial-model arms, but the evidence does not establish selective repair of the wrong model.** Pre-evidence errors, first-recipe divergence, pre-to-final error reductions, and frozen failure-aware selective-correction contrasts are reported by intervention locus. Every arm improves on average, but no locus passes the selective-correction criterion.

the exact law. Yet B2 exposed one nominal pair, no base coefficient or typed family/exponent, and an exact free-coefficient linear/power alias; a constant endpoint baseline reached MAE 0.00649 and the aligned positive control was only 1/5. B2 thus shows low error without stable exact-law expression on an underidentifying surface, not internal structural-identification failure (Fig.~3).

Matched DeepSeek/GPT contrasts were +0.0309/+0.0602 in A-P and +0.0645/+0.0915 in B2, with misspecified exact-law expression 0/5 in each model: a block-specific replication, not model ranking.

Changing only DeepSeek's reasoning effort within the Codex harness from high to low preserved all 15 post errors below 0.02 and misspecified expression at 0/5, while the contrast reversed to -0.0405 (2/5; $p = 0.8125$). This is same-harness robustness, not reasoning-off or configuration ranking.

Frozen B3 used reference-fitter-qualified multi-pair evidence and a disjoint roster. GPT completed 30/30 sessions with opaque/aligned/misspecified joint recovery 0/10, 5/10, and 0/10. DeepSeek retained 17 completions and 13 schema failures; failure-aware recovery and Top-1 were 0/30 and 0/30 versus 5/30 and 2/30 for GPT. Scheduled useful gain was 0/18 for both. Differential failures preclude ranking.

All 135 DeepSeek laws executed, but 84 lost information relative to final explicit predictions; mean law MAE was 0.237 and compression loss 0.069. Legal full-basis controls reproduced 135/135 prediction states at mean MAE 4.25×10^{-13}, localizing the gap to participant distillation rather than typed-interface capacity. Blind incumbent replay completed 726 executions for 121 cells, with recommendations better/equivalent/worse in 1/119/1.

The matched GPT-5.6-sol surface retained 126 completed, 3 failed, and 6 right-censored cells. All locus gates again failed. GPT-5.6-sol versus DeepSeek-v4-flash law MAE was 0.1753 (129 laws) versus 0.2371 (135 laws), and compression loss was 0.0142 versus 0.0686; blind gain was -0.0001

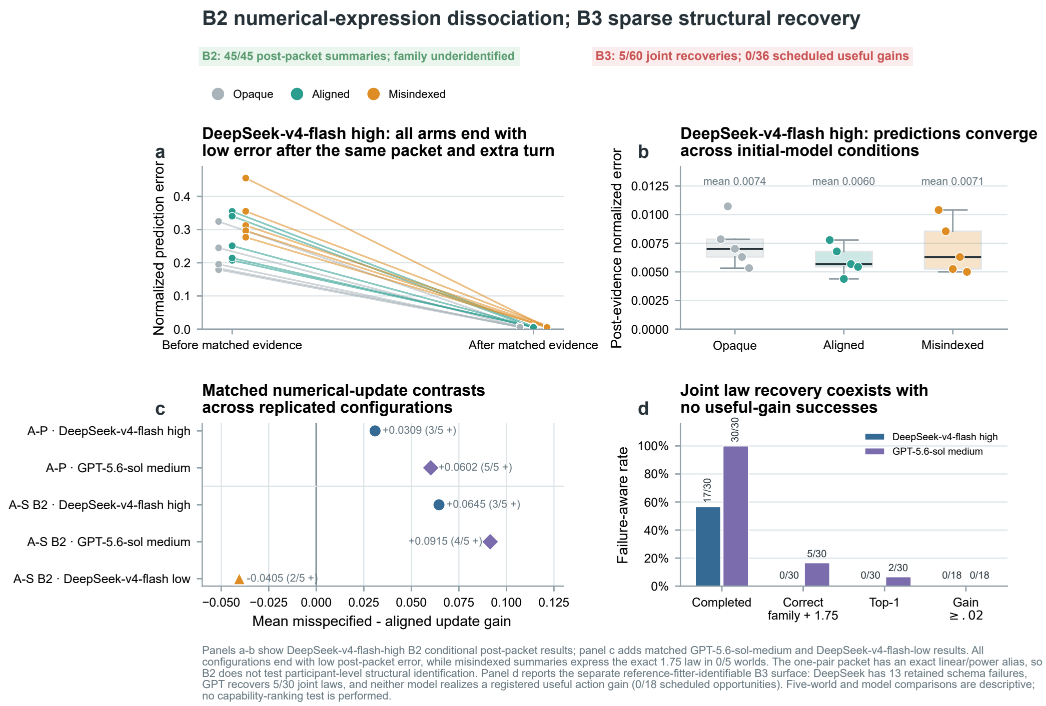

Figure 3: **B2 shows post-packet numerical–expression dissociation; identifiable B3 shows sparse structural recovery and no useful action gain.** Panels a–b detail DeepSeek-v4-flash-high conditional B2 results; the one-pair B2 surface underidentifies structural family. Panel c compares matched A-P DeepSeek-v4-flash/GPT-5.6-sol and B2 high/medium/low-reasoning contrasts. Panel d reports failure-aware completion, joint-law recovery, Top-1, and useful action gain on matched 30-cell B3 surfaces.

(126 cells) versus -0.0010 (121 cells). Lower observed compression error therefore coexisted with near-zero blind gain (Fig.~4); this matched cross-configuration comparison is descriptive, not a causal law-quality intervention.

5 Results: Law, Action, and Evaluation

5.1 Executable-law error is not a sufficient proxy for unseen action selection

The longitudinal DeepSeek matrix produced records for 45/45 scheduled cells. Truth evaluation and exact replay both completed 240/240 plan executions. Forty-two cells were uncontaminated and eligible for action metrics; two agent-induced resource/process failures and one session interruption in crystallization remain in the scheduled denominator.

Only 11/42 eligible terminal readouts selected the true Top-1 plan. Mean selected rank was 3.31 of 8 and mean normalized regret was 0.297. The uniform-random rank 4.5 is geometric context, not a no-evidence control, so these outcomes describe post-campaign competence rather than a causal benefit of exploration. Law and action also separated: 30 cells had an inadequate law and wrong action, 11 an inadequate law but correct action, one an adequate law but wrong action, and none an adequate law and correct action. Task means ranged from rank 2.00 for reaction safety to 4.58 for crystallization, and every pairwise arm contrast changed sign under some leave-one-cluster-out omission. We therefore make no pooled arm-level claim (Fig.~5a,b).

This separation survives continuous and threshold-sensitive analysis. Pooled law MAE correlated weakly with selected rank (Spearman $\rho = -0.073$, cluster-bootstrap 95% interval $[-0.380, 0.256]$) and normalized regret ($\rho = -0.133$, $[-0.452, 0.217]$), while task-specific rank associations reversed sign ($+0.524$, -0.592, and -0.007). Across law-MAE thresholds from 0.05 to 0.30, the adequate

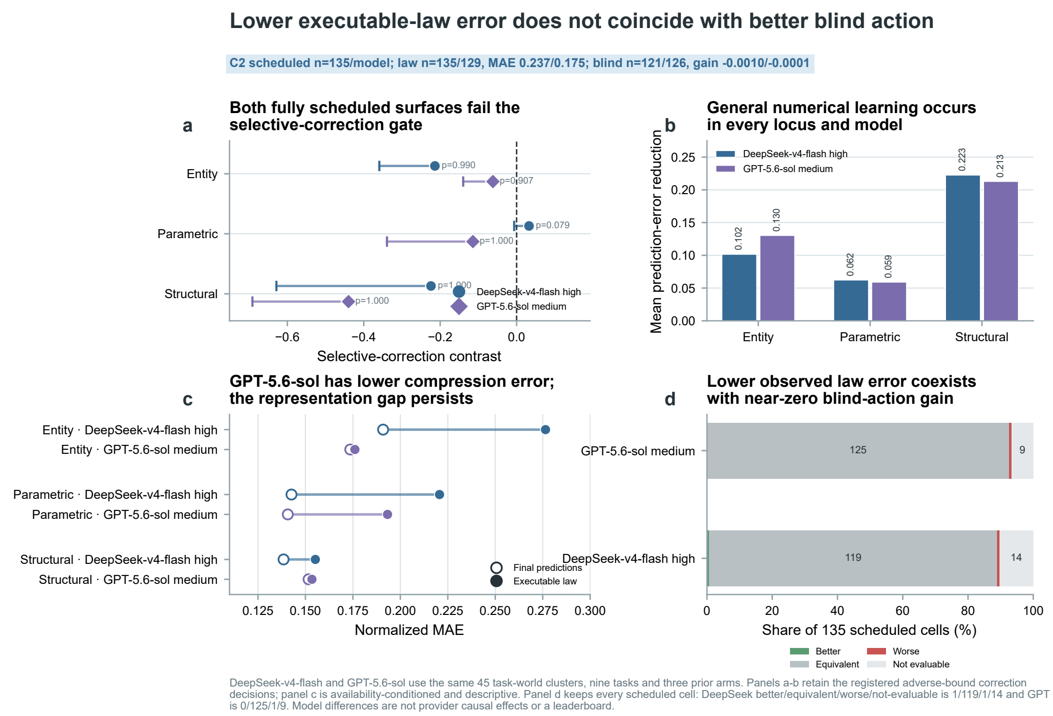

Figure 4: **Lower executable-law error does not coincide with better blind action. a,** Registered correction estimates and adverse lower bounds for both fully scheduled 135-cell C2 surfaces. **b,** Numerical learning by locus and model. **c,** Final-prediction versus executable-law MAE over 135 DeepSeek and 129 GPT laws. **d,** Blind outcomes over 121/126 evaluable cells while all 135/model remain scheduled; model contrasts are descriptive.

subset expanded from 1 to 34 cells but contained only 0 to 9 correct actions. The four-way table is therefore not a single-cutoff artifact: law error does not stably predict unseen-action quality across tasks. A decision-aligned reanalysis of this DeepSeek-v4-flash cohort then executed the last-available law from every frozen cell on the same eight frozen candidate plans. All 45 laws were evaluable, but their implied choices reached the true Top-1 in 0/45 cells, versus 11/45 when failures were retained for participant action. Among 42 cells with an action ranking, participants followed the law-implied Top-1 in only 12. Three cells without a terminal action ranking still retained an earlier executable law. Participant regret was lower than law-implied regret on average (0.344 versus 0.438), although the direction reversed in crystallization. Because neither law quality nor law following was randomized, this decomposes truth-law error from action-module utilization descriptively rather than estimating a causal law-to-action effect.

5.2 Four-condition strategies expose autonomy and learned-law-only limits

An independent successor scheduled four conditions over all 45 strata per model (360 slots). The all-scheduled failure-aware estimand retains donor, blocked-descendant and recipient failures. Autonomous-minus-no-evidence regret was -0.0913 for DeepSeek-v4-flash (95% task–world-cluster interval $[-0.2124, 0.0388]$) and +0.1102 for GPT-5.6-sol ($[-0.0533, 0.2794]$): directions differed and both intervals crossed zero. GPT's yoked- and learned-law-minus-none estimates were +0.2165 and +0.2459; these include missing donors and system failures, not pure evidence or artifact effects.

Autonomy also changed sign by task: DeepSeek -0.240/-0.406/+0.372 and GPT +0.089/-0.219/+0.460 for electrochemistry/safety/crystallization. Four registered contrasts are descriptive and unadjusted for multiplicity.

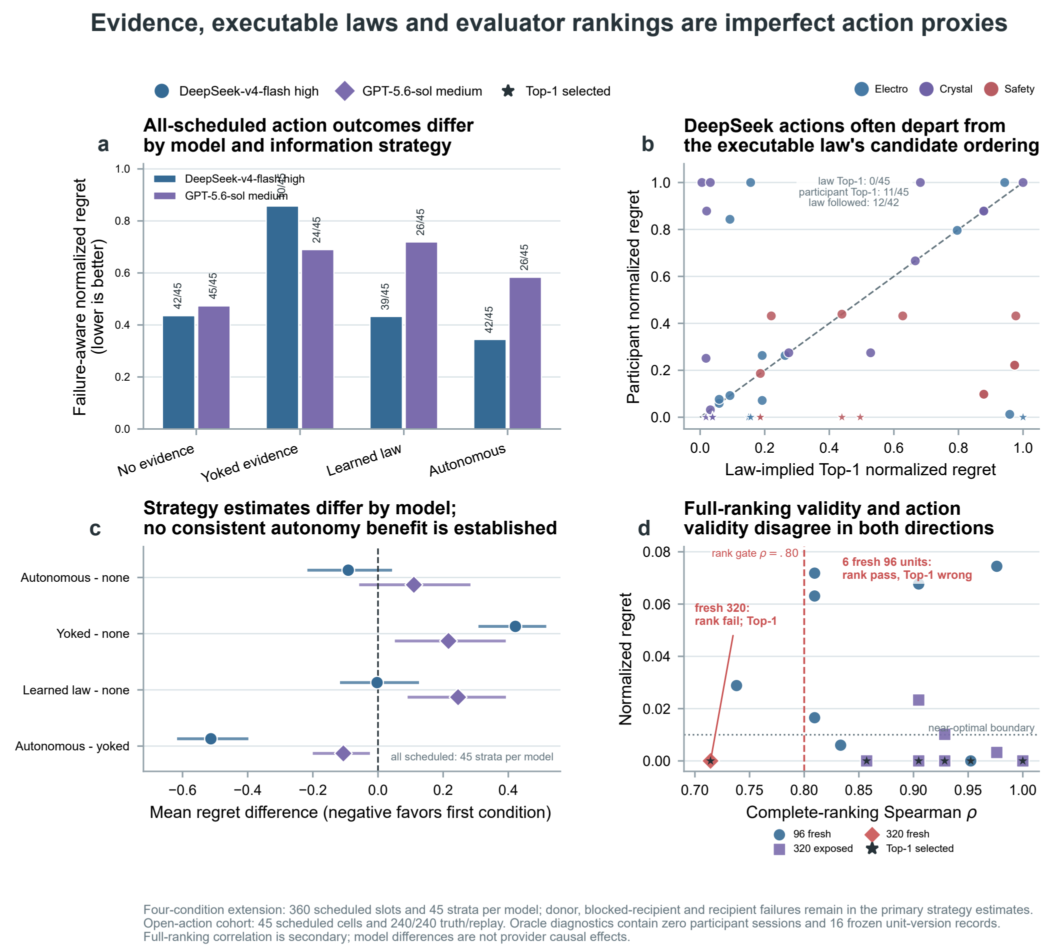

Figure 5: **Evidence, executable laws, and evaluator rankings are imperfect action proxies. a,c,** All-scheduled outcomes and clustered contrasts retain every donor and recipient failure. **b,** Last-available-law-implied versus participant regret for 45 frozen DeepSeek-v4-flash cells. **d,** Rank correlation versus regret for 16 frozen oracle unit-version records.

Donor-eligible autonomy contrasts (-0.1214/-0.1379; 42/26 strata) are post-treatment sensitivities; equal-task values are -0.0879/-0.0259. Yoked completion was 10/42 and 24/26. The block established no consistent autonomy or learned-law-only advantage (Fig.~5a,c).

5.3 Evaluator-level ranking and decision quality are different estimands

A planned five-condition follow-up required an outcome-disjoint oracle law to rank eight candidates with Spearman $\rho \geq 0.80$. The first eight fresh clusters completed 896/896 truth/replays, but the eighth obtained $\rho = 0.738095$ and the wrong Top-1; the frozen stop left all 225 participant sessions unstarted. Expanding 96 queries to a 320-query grid repaired 7/7 exposed units, yet the first fresh world failed at $\rho = 0.714286$ while selecting Top-1 with zero regret. A zero-execution diagnostic then reproduced all 16 unit versions: fresh 96-query rank gates passed 7/8 but Top-1 only 1/8, whereas the fresh 320-query unit was rank-fail/action-correct. Full-order correlation and decision validity therefore disagree in both directions (Fig.~5c,d; Table~4); the historical stops remain valid, while future controls should prioritize regret and near-optimality.

6 Related work

Self-driving laboratories and chemistry agents emphasize closed-loop execution, tool use, or endpoint optimization (Felton et al., 2021; Häse et al., 2021; Boiko et al., 2023; Bran et al., 2024). Virtual laboratories and process-control environments provide scalable interaction and safety (Beeler et al., 2024; Bloor et al., 2024; Malik et al., 2026; Chen et al., 2025). Scientific-discovery benchmarks increasingly test iterative experimentation, causal inference, and transferable knowledge (Jansen et al., 2024; Gandhi et al., 2025; Duan et al., 2025; Yang et al., 2026; Batzoglou, 2026). The published ChemWorld platform contributes programmable chemical worlds, transaction semantics and replay (Qiu et al., 2026); this work contributes the intervention and measurement programme for scientific-agent epistemics.

Model-discovery systems increasingly combine language models with Bayesian design, symbolic fitting or probabilistic programme search (Murphy, 2026; Wahl et al., 2026; Zheng et al., 2026). We instead ask whether evidence changes the right representation and whether that representation is usable for action. This follows the broader distinction between outcome and process validity (Ríos-García et al., 2026) and the warning that predictive success can coexist with underspecified or shortcut solutions (D'Amour et al., 2022; Geirhos et al., 2020). Causal mediation analysis would require identified interventions on intermediate representations (Imai et al., 2010); we do not infer such mediation from associated law and action readouts. The central contribution is therefore a tiered diagnostic design: controlled initial-model manipulations, conditional packet responses, failure-aware strategy estimates, descriptive law/action decomposition, and separate evaluator qualification.

7 Discussion and limitations

The systems search and learn numerically, yet lose information at distinct transitions: priors do not establish selective repair; underidentifying B2 accuracy does not certify structure; identifiable B3 recovery is sparse; and neither law error nor rank qualification suffices for action validity. The oracle stop prevented a decision-misaligned control from producing an uninterpretable participant effect.

7.1 A transition map for scientific-agent evaluation

Evaluation should therefore separate interventions from readouts, score prediction, law and action on their own units, and qualify controls on the intended endpoint. Regret, near-optimality and near-tie handling should be primary for selection, with full-rank metrics diagnostic. Evidence roles must also remain separate: the 96-query failure tests a control, the exposed 320-query pass tests construction, and the fresh failure tests generalization. Exact scheduled, eligible, completed and unstarted counts prevent these roles from being pooled into invented evidence.

7.2 Limitations and conclusion

This study covers two fixed simulated model–tool configurations, not provider ranking or laboratory fidelity. A-P/B2 have five worlds; B2 has an exact participant-visible alias and retrospective coding; DeepSeek low is not reasoning-off; and B3 retains 13 schema failures. Four-condition donor eligibility is sensitivity-only. Fixed execution order and earlier DeepSeek donors confound order with provider/time drift. These limits preclude ranking or mediation; context-reset portability and private replication remain untested. Scientific-agent claims must name interventions, units, failure rules and transfer boundaries.

AI USE STATEMENT

Generative AI tools assisted software implementation, experiment orchestration, data analysis, visualization code, literature discovery, and drafting and editing of the manuscript. The authors reviewed the study designs and frozen decision rules, inspected retained failures, verified reported denominators and claims against machine-readable records, and tested AI-assisted code before use. The authors take responsibility for the final content, including text, claims, code, and artifacts produced with generative-AI assistance.

REPRODUCIBILITY STATEMENT

The main paper defines the experimental units, interventions, estimands, failure handling, and stop boundaries. The appendix provides the execution and evaluation details needed to reconstruct each evidence layer. Sanitized scheduled-cell records, evaluator-derived metrics, replay and source digests, retained failure classifications, protocol projections, prompts, schemas, and figure source tables are included in the anonymous supplementary package.

ETHICS STATEMENT

The experiments use simulated chemical environments and no human participants, personal data, or live laboratory actuation. The principal risks are over-interpreting simulated results and treating an evaluator surrogate as the scientific objective; the claim boundaries and retained negative qualification results are reported to mitigate these risks.

REFERENCES

Serafim Batzoglou. ReplaySCM: A benchmark for executable causal mechanism induction from interventions, 2026. Preprint.

Chris Beeler, Sriram Ganapathi Subramanian, Kyle Sprague, et al. ChemGymRL: A customizable interactive framework for reinforcement learning for digital chemistry. *Digital Discovery*, 3: 742–758, 2024. doi: 10.1039/D3DD00183K.

Maximilian Bloor, José Torraca, Ilya Orson Sandoval, et al. PC-Gym: Benchmark environments for process control problems, 2024. Preprint, arXiv:2410.22093.

Daniil A. Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624:570–578, 2023. doi: 10.1038/s41586-023-06792-0.

Andres M. Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D. White, and Philippe Schwaller. Augmenting large language models with chemistry tools. *Nature Machine Intelligence*, 6:525–535, 2024. doi: 10.1038/s42256-024-00832-8.

Yimeng Chen, Piotr Piękos, Mateusz Ostaszewski, Firas Laakom, and Jürgen Schmidhuber. PhysGym: Benchmarking llms in interactive physics discovery with controlled priors. In *Advances in Neural Information Processing Systems*, 2025. URL https://openreview.net/forum?id=w8uII2qAmd.

Alexander D'Amour, Katherine Heller, Dan Moldovan, Ben Adlam, Babak Alipanahi, Alex Beutel, Christina Chen, Jonathan Deaton, Jacob Eisenstein, Matthew D. Hoffman, Farhad Hormozdiari, Neil Houlsby, Shaobo Hou, Ghassen Jerfel, Alan Karthikesalingam, Mario Lucic, Yian Ma, Cory McLean, Diana Mincu, Akinori Mitani, Andrea Montanari, Zachary Nado, Vivek Natarajan, Christopher Nielson, Thomas F. Osborne, Rajiv Raman, Kim Ramasamy, Rory Sayres, Jessica Schrouff, Martin Seneviratne, Shannon Sequeira, Harini Suresh, Victor Veitch, Max Vladymyrov, Xuezhi Wang, Kellie Webster, Steve Yadlowsky, Taedong Yun, Xiaohua Zhai, and D. Sculley. Underspecification presents challenges for credibility in modern machine learning. *Journal of Machine Learning Research*, 23(226):1–61, 2022. URL https://www.jmlr.org/papers/v23/20-1335.html.

Kourosh Darvish, Marta Skreta, Yuchi Zhao, et al. ORGANA: A robotic assistant for automated chemistry experimentation and characterization. *Matter*, 8(2), 2025. doi: 10.1016/j.matt.2024.10.015.

Haonan Duan, Stephen Zhewen Lu, Caitlin Fiona Harrigan, Nishkrit Desai, Jiarui Lu, Michał Koziarski, Leonardo Cotta, and Chris J. Maddison. Measuring scientific capabilities of language models with a systems biology dry lab, 2025. Preprint, arXiv:2507.02083.

Kobi C. Felton, Jan G. Rittig, and Alexei A. Lapkin. Summit: Benchmarking machine learning methods for reaction optimisation. *Chemistry–Methods*, 1:116–122, 2021. doi: 10.1002/cmtd.202000051.

Kanishk Gandhi, Michael Y. Li, Lyle Goodyear, Agam Bhatia, Louise Li, Aditi Bhaskar, Mohammed Zaman, and Noah D. Goodman. BoxingGym: Benchmarking progress in automated experimental design and model discovery, 2025. Preprint, arXiv:2501.01540.

Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2:665–673, 2020. doi: 10.1038/s42256-020-00257-z.

Florian Häse, Matteo Aldeghi, Riley J. Hickman, Loïc M. Roch, Melodie Christensen, Elena Liles, Jason E. Hein, and Alán Aspuru-Guzik. Olympus: a benchmarking framework for noisy optimization and experiment planning. *Machine Learning: Science and Technology*, 2:035021, 2021. doi: 10.1088/2632-2153/abedc8.

Kosuke Imai, Luke Keele, and Dustin Tingley. A general approach to causal mediation analysis. *Psychological Methods*, 15(4):309–334, 2010. doi: 10.1037/a0020761.

Peter Jansen, Marc-Alexandre Côté, Tushar Khot, Erin Bransom, Bhavana Dalvi Mishra, Bodhisattwa Prasad Majumder, Oyvind Tafjord, and Peter Clark. DiscoveryWorld: A virtual environment for developing and evaluating automated scientific discovery agents. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-0324.

Shreshth A. Malik, Tiarnan Doherty, Panagiotis Tigas, Muhammed Razzak, Stephen J. Roberts, Aron Walsh, and Yarin Gal. MADE: Benchmark environments for closed-loop materials discovery. In *International Conference on Machine Learning*, 2026.

Kevin Murphy. Model discovery agent: Llm-assisted bayesian experiment design for data-efficient discovery of mechanistic world models, 2026. Preprint, arXiv:2608.09696.

Jiangjie Qiu, Yijun Li, and Xiaonan Wang. ChemWorld: Programmable chemical worlds for controlled and replayable agent experimentation, 2026. Preprint, arXiv:2608.10792.

Martiño Ríos-García, Nawaf Alampara, Chandan Gupta, Indrajeet Mandal, Sajid Mannan, Ali Asghar Aghajani, N. M. Anoop Krishnan, and Kevin Maik Jablonka. Ai scientists produce results without reasoning scientifically, 2026. Preprint, arXiv:2604.18805.

Tao Song, Man Luo, Xiaolong Zhang, et al. A multiagent-driven robotic AI chemist enabling autonomous chemical research on demand. *Journal of the American Chemical Society*, 147(15):12534–12545, 2025. doi: 10.1021/jacs.4c17738.

Nathan J. Szymanski et al. An autonomous laboratory for the accelerated synthesis of inorganic materials. *Nature*, 624:86–91, 2023. doi: 10.1038/s41586-023-06734-w.

Aikaterini Vriza, Michael H. Prince, Tao Zhou, Henry Chan, and Mathew J. Cherukara. Operating advanced scientific instruments with AI agents that learn on the job. *npj Computational Materials*, 12:160, 2026. doi: 10.1038/s41524-026-02005-0.

Stefan Wahl, Raphaela Schenk, Ali Farnoud, Jakob H. Macke, and Daniel Gedon. A probabilistic framework for llm-based model discovery, 2026. Preprint, arXiv:2602.18266.

Junlin Yang, Dylan Zhang, Xiangchen Song, et al. CausaLab: A scalable environment for interactive causal discovery toward ai scientists, 2026. Preprint.

Tianshi Zheng, Kelvin Kiu-Wai Tam, Newt Hue-Nam K. Nguyen, et al. NewtonBench: Benchmarking generalizable scientific law discovery in llm agents. In *International Conference on Learning Representations*, 2026.

A Appendix

A.1 Environment semantics and intervention construction

An executable world is a stateful transition system with typed operations, measurement functions, finite resources, terminal assays, and an evaluator-only truth surface. A task publishes the operation and instrument contracts while withholding the latent state and registered relations. A scenario binds a world, public task description, resource card, and initial model. Initial-model arms alter one declared information locus without changing the transition process, measurement process, candidate actions, or evaluator truth.

The entity locus changes material identity or ontology. The parametric locus changes a quantitative response model while preserving the declared entities. The structural locus changes the functional form used to describe a mechanism. Aligned and misspecified models are matched in explicitness so that a contrast is not merely information versus no information. The opaque arm exposes the same laboratory interface but withholds the intervened model component. Exploratory, validation, prospective, and private world selections are disjoint by construction.

A.1.1 Task–locus roster and world splits

The nine prospective task–locus combinations are fixed before participant execution. The entity intervention swaps matched nominal descriptor rows while leaving physical identity fixed. Parametric and structural interventions replace only the declared response component; all other task text, tools, resources, queries, and world state remain matched.

Table 2: **Nine prospective task–locus combinations.** "Registered target" names the component manipulated in the initial model, not information revealed by the evaluator.

Locus	Task	Registered target	Held-out diagnostic
Entity	Electrochemical conversion	solvent identity descriptors	conversion/selectivity response
Entity	Reaction–crystallization	solvent identity descriptors	reaction and crystal outcomes
Entity	Reaction–distillation	solvent identity descriptors	reaction and separation outcomes
Entity	Partition discovery	extractant identity descriptors	phase-partition outcomes
Entity	Reaction safety	catalyst identity descriptors	yield, kinetics, and risk
Parametric	Electrochemical conversion	quantitative potential-response direction	peak-and-collapse profile
Parametric	Reaction safety	quantitative catalyst/thermal response	productive versus hazardous regime
Structural	Partition discovery	linear versus power-form phase relation	registered exponent 1.75
Structural	Reaction–crystallization	pathway/topology and accumulation form	multistage outcome surface

Development and qualification use seeds 0–4 and never enter the prospective denominator. Public worlds are five task-specific SHA-256-derived identities per task; their values are fixed in the released configuration. The private split is bound by a commitment but was not unsealed or executed. Matched-evidence and action studies use separately frozen public surfaces. Sessions sharing a task–world but differing in prior arm are repeated measurements within one inference cluster.

A.1.2 Metrics and registered correction gates

For prediction or executable law f, truth y, registered query–metric pairs $j = 1, \dots, J$, and fixed positive metric scale s_j (one for bounded metrics unless overridden), normalized error is

$$E(f) = \frac{1}{J}\sum_{j=1}^{J} \min\left(\frac{|f_j - y_j|}{s_j}, 1\right).$$

For candidate scores q_i, selected candidate a, best $q_{\max}$, and worst $q_{\min}$, normalized regret is $(q_{\max} - q_a)/(q_{\max} - q_{\min})$, with zero assigned when the packet range is numerically zero. A missing or invalid terminal ranking receives failure-aware regret one and Top-1 zero. Near-optimal selection means raw regret at most 0.01; pairwise ordering excludes truth pairs separated by less than 0.01.

Let $I_a = E_{a,0} - E_{a,K}$ and $C = I_{\text{mis}} - I_{\text{aligned}}$. The entity gate is a three-component intersection–union test at one-sided $\alpha = 0.05$: the lower bound for C must exceed zero, misspecified improvement must exceed zero, and aligned improvement must exceed the non-inferiority margin -0.05. Parametric and structural gates use equal-task fixed-effect estimates of C, require the one-sided lower bound above zero and positive direction in every included task. For an incomplete cell, confirmatory bounds use the adverse interval $[-1, +1]$ for arm improvement; the corresponding contrast can therefore span $[-2, +2]$. Last-observation-carried-forward and zero improvement are observed-point sensitivities only and do not generate the confirmatory p values.

A.2 Persistent execution and checkpoints

Each participant cell is one persistent agent process. At every turn, the agent may call only the public laboratory tools. The host validates schemas and preconditions, applies resource accounting, commits successful transactions, and returns public observations. It does not choose or repair an experiment on the participant's behalf. Entity, parametric, and structural campaigns permit eight, ten, and twelve complete experiments, respectively. Checkpoints occur at experiments 0/2/4/6/8, 0/2/4/7/10, and 0/3/6/9/12.

Every checkpoint records model-reliability judgements, query predictions, uncertainty, public evidence references, a typed executable law, and the intended next experiment. Checkpoint repair is part of the evaluated interface burden. All eventually yielded valid records, but rejected attempts remain in the process log and resource accounting.

A.2.1 System, prompt, tool, and budget contract

`DeepSeek-v4-flash` with `reasoning_effort=high` supplies the primary DeepSeek C2, matched-evidence, B3-successor, and action surfaces; the same model at `low` supplies only the completed B2 reasoning-budget ablation. `GPT-5.6-sol` with `reasoning_effort=medium` supplies the matched GPT replications. Both run through the Codex session harness; "Codex" therefore denotes execution infrastructure, not a third participant model. Provider aliases, run dates, receipt counts, and usage fields remain in the retained internal records rather than the anonymous supplement. The supplement retains sanitized model/configuration labels, scheduled completion denominators, and derived resource summaries. Temperature is not exposed or set by the harness. Request timeout is 1,200 s; finalization timeout is 600 s; progress is emitted every 30 s.

Each C2 cell receives one task/scenario prompt, the same public operation schemas, the arm-specific initial model, and no evaluator truth. A persistent transcript spans the complete campaign. The fixed 8/10/12-experiment budgets allow at most 96/110/168 operation attempts, 480k/600k/720k uncached input tokens, 128k/160k/192k output tokens, and 7,200/7,200/9,000 s wall time, respectively. Each cell uses at most two provider calls within the same session; infrastructure-only resume is capped once, while any persisted scientific trajectory forbids replacement. Matched-evidence assays use exactly two same-thread turns: a pre-evidence submission and a post-packet submission. The supplement includes the exact static system prompts, selected prompt-generating function sources, public protocol projections, representative strict response schemas, and the executable-law contract. The package also contains all 45 sanitized B2 public summaries, the retrospective expression-coding function, three-configuration all-arm counts, and the participant-visible B2 identifiability audit.

A.2.2 Executable-law schema and ActionPlan construction

A law contains one metric equation for every required metric, an intercept, identity or logistic link, output bounds, evidence references, confidence, applicability, and limitations. Each metric permits at most 64 terms drawn from linear, quadratic, cubic, pairwise interaction, categorical indicator, and category-conditional linear/quadratic/cubic bases. Terms may reference only the task's registered feature IDs; evidence references may name only participant-visible assays. The production parser and executor score the law directly, so prose that cannot be parsed does not silently receive a law score.

Terminal candidates are eight complete, compiler-valid ActionPlans selected from a registered 16-query public-feature roster without reading outcomes. Each packet declares every ordered operation and parameter, fresh initial state, measurement positions, terminal assay, omitted optional operation, and objective. The other eight rows form the disjoint prediction/oracle-fit roster. Public-plan, truth-plan, and replay hashes must agree exactly. Candidate outcomes and evaluator ranks remain hidden until scoring.

A.3 Evaluator-owned prediction and law assays

Entity checkpoints contain four prespecified counterfactual queries; parametric and structural checkpoints contain sixteen. The evaluator executes each unique task–world query independently, never returns its truth to the participant, and shares truth across prior arms and checkpoints. The primary readout is mean normalized absolute error across registered query–metric pairs. The executed cohort contains 420/420 truth executions, 1,620 query–metric truth values, 675/675 checkpoints, 6,300 predictions, and 24,300 scored values.

Final typed laws are executed on the same registered coordinates. Schema validity, coordinate coverage, truth-normalized error, change from pre-evidence and effective-final checkpoints, and consistency with explicit final predictions are retained separately. This assay establishes executability and in-domain compression only. Reusability would require new coordinates or transfer to a context-reset process.

The matched GPT-5.6-sol evaluator uses the identical 45-cluster, nine-task, three-arm C2 surface. DeepSeek-v4-flash and GPT-5.6-sol have 121/135 and 126/135 completed cells. Their exact evaluator denominators are 675/675 versus 669/675 scored checkpoints, 135/135 versus 129/135 executable laws, and 121/135 versus 126/135 cells with evaluable blind gain. Overall mean prediction improvement is 0.1198/0.1329, final prediction error is 0.1685/0.1614, law MAE is 0.2371/0.1753, law compression loss is 0.0686/0.0142, and blind gain is -0.0010/-0.0001. GPT-minus-DeepSeek paired descriptive differences are +0.0131 for prediction improvement, -0.0071 for final error, -0.0648 for law MAE, and -0.0579 for compression loss. Both models fail all three registered selective-correction gates.

A.4 Typed-law capacity and distillation controls

To distinguish participant compression failure from an underpowered output schema, we refitted each of the 135 complete final prediction vectors with legal identity-link laws over the registered feature coordinates and allowed basis functions. A full-basis fit used up to 64 terms per metric. A term-matched fit used exactly the participant's submitted term budget for each metric. A leave-one-query-out fit withheld each registered query in turn and predicted it from all remaining queries. Every fitted law was parsed and executed by the production law evaluator; independently computed design-matrix predictions had to agree with executor output within 10^{-10}.

The participant laws differed from their own final predictions by mean MAE 0.1539. Full-basis fits reproduced 135/135 final prediction states with mean MAE 4.25×10^{-13}; term-matched fits reduced the error to 0.0114 and leave-one-query-out fits reached 0.0788. The control uses the same registered coordinate domain and therefore tests representation capacity and distillation, not global mechanistic identification or transfer.

A.5 Failure-aware inference

The unit of prospective inference is the independently selected task–world cluster, not a checkpoint, query, experiment, or replay. The three prior arms within a cluster are matched. Locus-specific selective-correction contrasts are computed first and are not pooled across different intervention semantics. Failed scientific cells remain assigned. Confirmatory correction bounds assign an incomplete post-outcome the adverse improvement range $[-1, +1]$; last-observation-carried-forward and zero improvement are observed-point sensitivities only. Infrastructure recovery is allowed only when no scientific trajectory has persisted and only under the fixed attempt limit.

The matched-evidence B2 contrast has five independent worlds. Its interval and exact one-sided sign-flip result are descriptive because the block was designed to localize a transition, not to establish a population-level arm effect. Recipe divergence is likewise a manipulation check: without repeated same-arm sessions, provider stochasticity cannot be separated from intervention sensitivity at exact-recipe resolution.

DeepSeek and GPT-5.6-sol medium used identical matched-evidence worlds, arms, evidence packets, queries, and scoring rules. Each completed 15/15 A-P and 15/15 A-S B2 formal sessions with no failed cells; canaries were excluded. A-P misspecified-minus-aligned update contrasts were 0.0309 and 0.0602, with 3/5 and 5/5 positive worlds. A-S B2 contrasts were 0.0645 and 0.0915, with 3/5 and

4/5 positive worlds; misspecified exact 1.75-law expression was 0/5 in both models. Configurations are reported separately, and no model-superiority test is performed.

The B2 exact-law counts are retrospective keyword coding of public `model_summary` and `evidence_assessment` fields; no typed family or exponent field was preregistered. A later participant-visible audit found that evidence and scoring fixed one nominal solvent/extractant pair, did not expose the base partition coefficient, and admitted an exact alias between the 1.75-power law and a free-coefficient linear law (coefficient multiplier 3.13588). A constant endpoint baseline obtained mean scoring MAE 0.00649, while the aligned DeepSeek-high exact-law positive control was only 1/5 and failed its readout criterion. B2 therefore supports conditional low post-packet error and lack of stable exact-law expression on an underidentifying surface, not a participant-level structural-identification failure. The separate B3 control below supplies that identifiable test.

The reasoning-budget ablation kept the DeepSeek model, Codex harness, prompts, schemas, public packets, worlds, and evaluator fixed while changing only `reasoning_effort` from high to low. It is not provider-level thinking-off, which would require a different direct-controller harness. The B2 interface canary passed 3/3 and the formal block completed 15/15 sessions, 30/30 turns, 15/15 same-thread continuations, and 360/360 pre plus 360/360 post scoring terms, with no failed session or infrastructure predecessor. Mean post errors were 0.0067/0.0069/0.0069 for opaque/aligned/misspecified. The registered contrast was -0.0405, with 2/5 positive worlds, exact one-sided $p = 0.8125$, and descriptive interval [-0.1559, 0.0749]. Misspecified exact 1.75-law expression remained 0/5; all five misspecified post errors were at most 0.02. Aligned exact-law expression was 1/5 for DeepSeek high, 0/5 for GPT medium, and 2/5 for DeepSeek low, reinforcing that the free-text expression readout itself lacks a strong positive control.

Provider-reported DeepSeek reasoning output was 506,637 tokens at high effort and 400,639 at low effort on the matched B2 block, a 20.9% reduction. These resource fields are descriptive within the DeepSeek provider and are not compared with GPT accounting. The separate low-effort A-P canary was platform-defective: progress events existed, but terminal cell receipts and a canary summary did not. It is retained as an unscored partial; A-P low formal execution remained 0/15 and supplies no effect estimate.

A.6 Reference-fitter-identifiable law and action control

The independent B3 control used five frozen public structural-partition worlds, the same opaque, aligned, and misspecified prior arms, and two independent fresh GPT-5.6-sol medium sessions per arm and world. Provider-free development qualification selected eight evidence rows spanning four nominal pairs and a disjoint eight-query scoring/action roster. Public truth was executed before any participant call. All five worlds entered family, exponent, prediction, Top-1, rank, and regret denominators; action gain was scored only in the three worlds whose best candidate exceeded the visible evidence incumbent by at least 0.02. The evidence and scoring rosters, five worlds, three arms, two replicates, 0.10 exponent tolerance, and 0.02 opportunity threshold were frozen before participant execution.

A three-session canary tested only two-turn schema validity, packet identity, same-thread continuity, typed law fields, the exact scoring denominator, and selection of an unexecuted candidate. It passed 3/3 and did not enter the formal scientific denominator. The full 30-cell block then completed 30/30 with no failed cells or replacement. All sessions preserved same-thread continuity and returned 60/60 completed formal turn receipts. The canary and formal blocks together used 33 provider session attempts, 66/66 completed turn receipts, zero retries, zero tool events, and zero participant physical experiments. One formal post turn recorded a transient provider error event before returning a completed receipt; the event remains in the resource ledger and the cell remains completed.

Joint structural recovery required both the registered power family and exponent error at most 0.10. Counts for opaque, aligned, and misspecified arms were 0/10, 5/10, and 0/10. The corresponding mean post-evidence errors were 0.0367, 0.0215, and 0.0378. The aligned world-mean exponent error was lower than each comparator in all five worlds. Top-1 counts were 0/10, 2/10, and 0/10; both Top-1 choices came from one action-ineligible world. None of the 18 action-eligible cells reached gain 0.02, including neither of the two action-eligible cells with joint structural recovery. A matched DeepSeek B3 successor then started from its first cell on the same scientific surface and retained every scheduled outcome. It completed 17/30 cells; 13/30 were participant-schema failures.

Failure-aware joint recovery was 0/30, Top-1 was 0/30, and mean regret was 0.9579; completed-cell post MAE was 0.0928. The corresponding GPT-5.6-sol values were 5/30, 2/30, 0.7594, and 0.0320. Useful-gain success was 0/13 versus 0/18 among completed evaluable opportunities and 0/18 for both models on the fixed scheduled-opportunity denominator, with failures and unavailable rows counted as zero. The matched comparison is descriptive because model configuration was not randomized and schema-failure patterns differed.

A.7 Cross-model successor denominators

Historical stopped partials and stop boundaries remain immutable. They are not continued, pooled, or converted into successful cells. Each successor instead starts at its first scheduled unit with a complete failure-aware denominator.

The GPT-5.6-sol C2 successor scheduled all 135 task–world–prior cells and terminated with 126 completed, 3 failed, and 6 right-censored sessions, completing 1,253/1,260 participant experiments. Its provider-free evaluator completed 420/420 truth executions, scored 669/675 checkpoints and 129/135 laws, and launched 756/810 scheduled blind executions; 54 remained unstarted. The matched 135-cell DeepSeek/GPT analysis uses task–world cluster bootstrap intervals and never interprets model differences as provider effects.

The DeepSeek B3 successor similarly schedules 30 cells and retains 17 completed plus 13 participant-schema failures. Its historical canary is excluded rather than spliced into the new denominator. The GPT B3 cohort remains the independent completed 30-cell block described above.

The four-condition action successor schedules 45 strata by four conditions for each model: 180 slots per model and 360 total. DeepSeek reuses 45 immutable autonomous donors, of which 42 are eligible; GPT-5.6-sol creates a separate 45-donor cohort, of which 26 are eligible. No-evidence is scheduled for all strata, while yoked and learned-law recipients require the corresponding donor. Missing-donor descendants remain `not_started_due_to_missing_donor`. The primary analysis is the model-specific all-scheduled failure-aware strategy estimand over all 45 strata. DeepSeek/GPT autonomy-minus-no-evidence estimates are -0.0913 and +0.1102 with clustered intervals [-0.2124, 0.0388] and [-0.0533, 0.2794]. Donor-eligible estimates (-0.1214/-0.1379) are availability-conditioned sensitivities because eligibility is post-treatment; equal-task sensitivities are -0.0879 for DeepSeek and -0.0259 for GPT. The cross-model common donor-eligible population contains 26 strata from 13 task–world clusters. Yoked completion is 10/42 for DeepSeek and 24/26 for GPT, so autonomous–yoked contrasts mix scientific action quality with recipient-system failure and are not pure experiment-selection effects. The autonomous-minus-no-evidence task estimates were -0.2398/-0.4060/+0.3720 for DeepSeek and +0.0891/-0.2191/+0.4605 for GPT over electrochemistry/safety/crystallization, respectively. Thus the pooled direction masks strong task heterogeneity. All four registered contrasts and their bootstrap intervals are descriptive and were not multiplicity-adjusted. Conditions followed fixed manifest and dependency order rather than randomization or counterbalancing; DeepSeek autonomous donors also predated their recipients. Provider/time drift and condition-order effects are therefore not separable.

A.7.1 Recovery and composite provenance

Platform repair never changed a scientific question, coverage set, metric, or gate after observing an outcome. Affected blocks restarted from their first affected unit; discarded or superseded records remain in the failure timeline and do not enter the current estimate.

Table 3: **Defect and recovery provenance for current publication surfaces.**

Surface	Retained incident	Recovery boundary	Current inclusion
DeepSeek C2	one 15-cell shard affected by runtime/evaluator defects	whole shard restarted; evaluator rerun over the full composite	120 unaffected cells plus one 15-cell replacement shard
GPT C2	no inherited stopped cell	new 135-cell block from cell 1	all 135 scheduled outcomes
DeepSeek B3	historical three-cell canary failed schema	independent 30-cell successor from cell 1	17 complete plus 13 schema failures
Four-condition yoked	duplicate field deletion caused recipient KeyError	entire yoked condition restarted from its first admitted unit	recovered condition only; original failures retained in timeline
Oracle grids	path-length and evidence-reference limits	identical scientific block restarted after platform repair	only complete version-specific qualification blocks

The four-condition report remains a prospective development experiment rather than confirmatory evidence; the reanalysis does not rename it formal evidence. The DeepSeek C2 current composite is reported as a composite, not as one uninterrupted launch. Exact source bindings and excluded incident counts are in the supplement.

A.8 Complete-plan action assay

In each open-action cell, a persistent agent performs twelve autonomous experiments and commits checkpoints after 0, 3, 6, 9, and 12 experiments. The host then reveals eight unseen ActionPlans. Each plan specifies ordered operations, all submitted parameters, initial-state assumptions, intermediate measurement positions, terminal assay, and declared omitted operations. Candidate outcomes, ranks, and other-arm evidence stay hidden. The evaluator verifies identity of disclosed, truth-evaluated, and replayed plans.

The primary endpoint is normalized regret of the selected plan within its task–world candidate set. Selected rank, Top-1, pairwise ordering, and law adequacy are secondary or diagnostic. Forty-five cells were scheduled across three tasks, five worlds, and three initial-model arms. Forty-two were eligible. The two agent-induced resource/process failures and one provider interruption are retained as failures in the scheduled denominator and are not replaced. An independent repair is reported only as sensitivity evidence because it follows a new trajectory.

The continuous law–action analysis used only those same 42 originally eligible cells; the three failures stayed in the 45-cell scheduled denominator and were not imputed. Pearson and Spearman associations were computed pooled and by task. Uncertainty resampled the 15 frozen task–world clusters, retaining all eligible arms within a sampled cluster, with 10,000 bootstrap replicates and seed 20260827. The pooled Spearman coefficients were -0.073 for law MAE versus selected rank and -0.133 for law MAE versus normalized regret, with cluster-bootstrap 95% intervals $[-0.380, 0.256]$ and $[-0.452, 0.217]$. A prespecified sensitivity table swept law-MAE thresholds 0.05, 0.075, 0.10, 0.15, 0.20, 0.25, and 0.30 without selecting a cutoff from action outcomes.

A decision-aligned reanalysis used the last-available executable law from all 45 frozen DeepSeek-v4-flash longitudinal cells and reconstructed the same eight candidate feature packets without new participant, truth, or physics execution. Three cells had no terminal action ranking but retained an earlier executable law. For each law, truth-law error is the normalized regret of its implied Top-1. Action-utilization delta is participant regret minus law-implied regret. All 45 laws executed; none implied the true Top-1, whereas participant action selected Top-1 in 11/45 scheduled cells. Among 42 cells with a valid participant ranking, the participant followed the law-implied Top-1 in 12. Mean law-implied and participant regret were 0.438 and 0.344. This is a descriptive decomposition: neither law quality nor law use was randomized.

A.9 Oracle-control qualification

The unexecuted causal follow-up crossed three tasks, five worlds, three priors, and five conditions: no evidence, stepwise yoked evidence, autonomous exploration, learned law in a fresh context, and a provider-free oracle law. The planned denominator was 225 participant sessions, 45 autonomous donors, and 540 donor experiments. Participant execution required every fresh task–world oracle to pass candidate opportunity checks and to attain Spearman rank correlation at least 0.80 over the eight candidate plans using fit data disjoint from candidate outcomes.

The 96-query qualification stopped after eight of fifteen clusters. It completed 896/896 truth executions and exact replays, with eight candidate gates and seven oracle rank gates passing. The retained crystallization failure had $\rho = 0.738095$, Top-1 disagreement, and zero fit/candidate overlap. No operational canary or participant session began.

This failure initiated a transparent development sequence rather than a retrospective threshold change. Version 0.2 first failed on a fresh electrochemical world at $\rho = 0.785714$. Version 0.3 added exact typed-law distillation but failed on a fresh crystallization world at $\rho = 0.595238$. Version 0.4 fixed an ExtraTrees predictor: it replayed all 25 exposed construction worlds successfully, then stopped after five fresh units when the fifth reached only $\rho = 0.785714$ (four pass, one fail, ten unstarted). Each version used a newly held-out fresh surface, stopped at its first registered failure, read no candidate outcomes during fitting, and made zero provider calls. These attempts are not pooled into a success rate and do not modify the original stop decision.

The 320-query construction held the fitted ExtraTrees family fixed while expanding coverage to 64 global and 256 candidate-neighborhood queries. A platform-defective partial caused by an evidence-reference schema limit was retained and not reused. After the contract repair, an independent construction run completed 2,352/2,352 truth executions and replays across seven exposed units, passing all seven and repairing four historical failures. The first prospective world then completed 336/336 executions and replays but failed at $\rho = 0.714286$; the remaining fourteen were not started. Candidate outcomes were never read during fitting.

The gate-alignment analysis added no execution. It re-read sixteen frozen unit versions and exactly reproduced their original Spearman and Top-1 values. Its action endpoints were Top-1, normalized regret, selection within 0.01 of optimum, and near-tie-aware pair ordering. It did not alter any historical result, threshold, or stop decision.

For eight candidates without ties, $\rho = 1 - \sum_i d_i^2/84$. Thus the retained 96-query failure at $\rho = 0.738095$ has $\sum_i d_i^2 = 22$, whereas the fresh 320-query failure at $\rho = 0.714286$ has 24. The scalar measures total displacement, not whether the best action moved.

Table 4: **Complete-ranking gates and action endpoints are different estimands.**

Frozen block	n	Rank pass	Top-1	≤ 0.01 regret
Fresh 96-query	8	7	1	3
Exposed 320-query	7	7	4	6
Fresh 320-query	1	0	1	1

A.10 Additional scope boundaries

The study covers simulated chemistry and two fixed agent–tool configurations, not laboratory fidelity or universal chemical coverage. C2 and B3 have matched scheduled surfaces; A-P/B2 have complete separate denominators plus a B2-only DeepSeek-low ablation. Differential failures and non-random provider assignment prohibit model ordering. The public cohort is not private confirmation, and high-fidelity artifact portability remains untested. The autonomous open-action cohort lacks a same-agent no-evidence baseline; the oracle-free successor adds four information strategies, but donor/recipient failures limit mediation claims. The original five-condition oracle cohort remains unexecuted.